Stefano Manzini
Laboratory of Experimental Pharmacology and Integrative Biology of Atherosclerosis

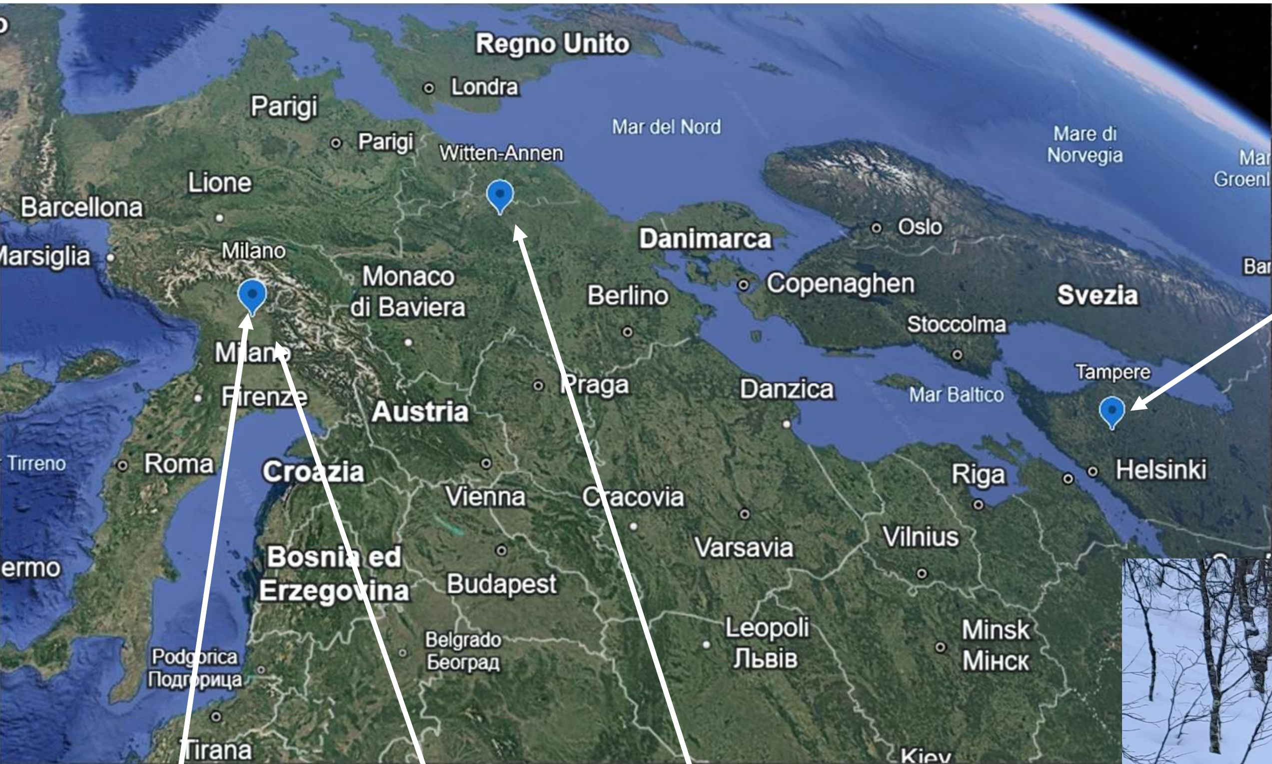
RNAseq

a little of
History, Theory, Practice

UNIVERSITÀ DEGLI STUDI DI MILANO

Dipartimento di Eccellenza MUR
2023-2027

Dipartimento di Scienze Farmacologiche e Biomolecolari Rodolfo Paoletti



Tampereen yliopisto
Tampereen ammattikorkeakoulu

UNIVERSITÀ DEGLI STUDI DI MILANO

UNIVERSITÀ DEGLI STUDI DI MILANO BICOCCA

universität Witten/Herdecke

737-738 (25 April 1953) | doi:10.1038

Molecular Structure of Nucleic Acids: A Structure for Deoxyribose Nucleic Acid

J. D. WATSON & F. H. C. CRICK

Medical Research Council Unit for the

1953!

EARLY DAYS OF DNA SEQUENCING

Fifteen years elapsed between the discovery of the DNA double helix and the first experimental determination of a DNA sequence. This delay was caused by several factors that made the problem intimidating:

(i) The chemical properties of different DNA molecules were so similar that it appeared difficult to separate them.
(ii) The chain length of naturally occurring DNA molecules was much greater than for proteins and made complete sequencing seems unapproachable.
(iii) The 20 amino acid residues found in proteins have widely varying properties that had proven useful in the separation of peptides. The existence of only four bases in DNA therefore seemed to make sequencing a more difficult problem for DNA than for protein.
(iv) No base-specific DNAases were known. Protein sequencing had depended upon proteases that cleave adjacent to certain amino acids.

DNA sequencing: bench to bedside and beyond†

Clyde A. Hutchison III*

J. Craig Venter Institute, 9704 Medical Center Drive, Rockville, MD 20850, USA

Nucleic Acids Research, 2007, Vol. 35, No. 18 ***6227–6237***

doi:10.1093/nar/gkm688

1977

Proc. Natl. Acad. Sci. USA
Vol. 74, No. 2, pp. 560–564, February 1977
Biochemistry

A new method for sequencing DNA

(DNA chemistry/dimethyl sulfate cleavage/hydrazine/piperidine)

ALLAN M. MAXAM AND WALTER GILBERT

Department of Biochemistry and Molecular Biology, Harvard University, Cambridge, Massachusetts 02138

Contributed by Walter Gilbert, December 9, 1976

ABSTRACT DNA can be sequenced by a chemical

Proc. Natl. Acad. Sci. USA
Vol. 74, No. 12, pp. 5463–5467, December 1977
Biochemistry

DNA sequencing with chain-terminating inhibitors

(DNA polymerase/nucleotide sequences/bacteriophage ϕX174)

F. SANGER, S. NICKLEN, AND A. R. COULSON

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THE SPECIFIC CHEMISTRY

A Guanine/Adenine Cleavage (2). Dimethyl sulfate methylates the guanines in DNA at the N7 position and the adenines at the N3 (3). The glycosidic bond of a methylated purine is unstable (3, 4) and breaks easily on heating at neutral pH, leaving the sugar free. Treatment with 0.1 M alkali at 90° then will cleave the sugar from the neighboring phosphate groups. When the resulting end-labeled fragments are resolved on a polyacrylamide gel, the autoradiograph contains a pattern of dark and light bands. The dark bands arise from breakage at guanines, which methylate 5-fold faster than adenines (3).

This strong guanine/weak adenine pattern contains almost half the information necessary for sequencing; however, ambiguities can arise in the interpretation of this pattern because the intensity of isolated bands is not easy to assess. To determine the bases we compare the information contained in this column of the gel with that in a parallel column in which the breakage at the guanines is suppressed, leaving the adenines apparently enhanced.

An Adenine-Enhanced Cleavage. The glycosidic bond of methylated adenosine is less stable than that of methylated guanosine (4); thus, gentle treatment with dilute acid releases adenines preferentially. Subsequent cleavage with alkali then produces a pattern of dark bands corresponding to adenines with light bands at guanines.

Cleavage at Cytosines and Thymines. Hydrazine reacts with thymine and cytosine, cleaving the base and leaving ribosylurea (5–7). Hydrazine then may react further to produce a hydrazone (5). After a partial hydrazinolysis in 15–18 M aqueous hydrazine at 20°, the DNA is cleaved with 0.5 M piperidine. This cyclic secondary amine, as the free base, displaces all the products of the hydrazine reaction from the sugars and catalyzes the β-elimination of the phosphates. The final pattern contains bands of similar intensity from the cleavages at cytosines and thymines.

Cleavage at Cytosine. The presence of 2 M NaCl preferentially supresses the reaction of thymines with hydrazine. Then, the piperidine breakage produces bands only from cytosine.

Dimethyl sulfate + slight alkali treatment: G + A

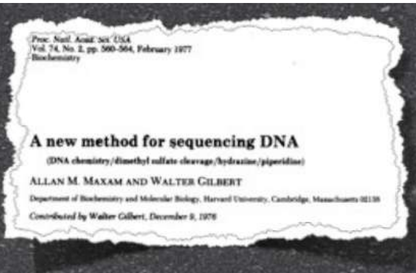
Dimethyl sulfate + diluted acid: A

Hydrazine + piperidine T + C

Hydrazine + NaCl + piperidine: C

pyrimidines purines

Chemical DNA sequencing



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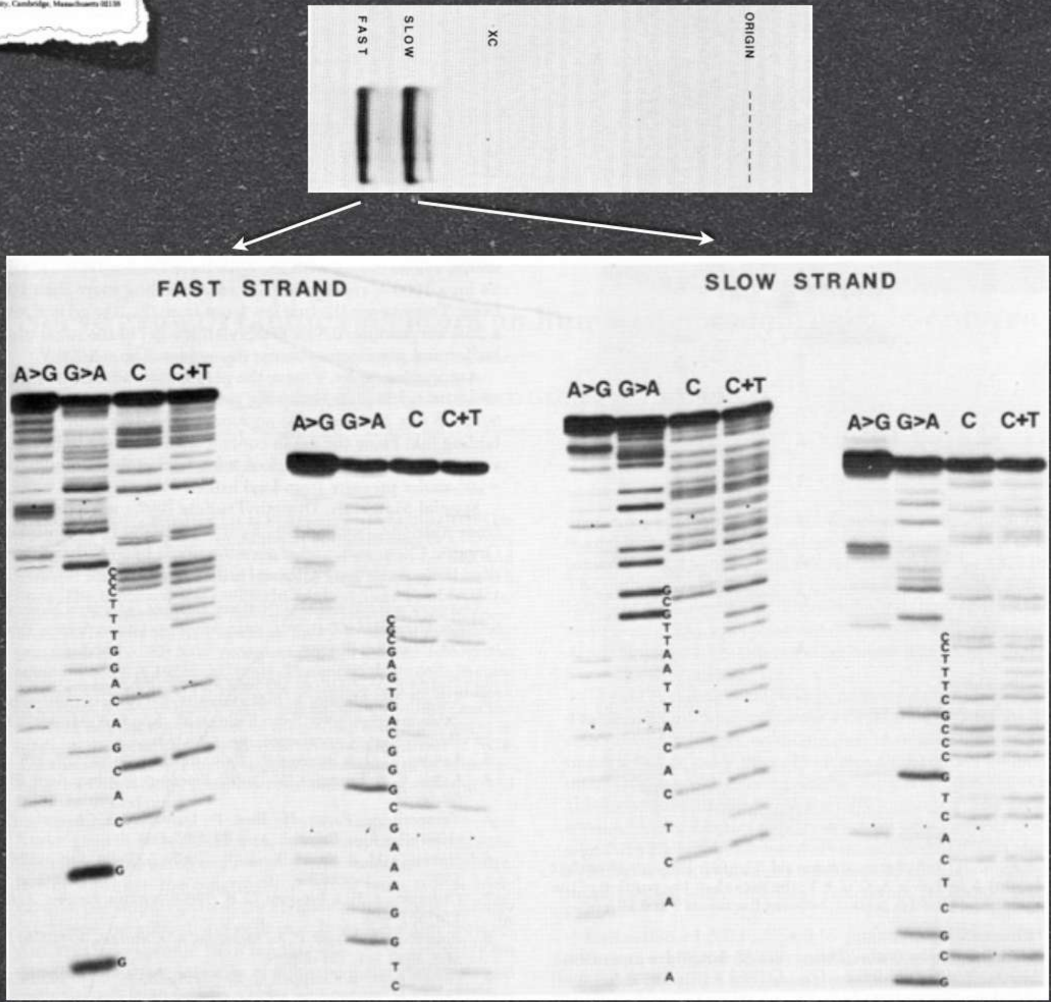
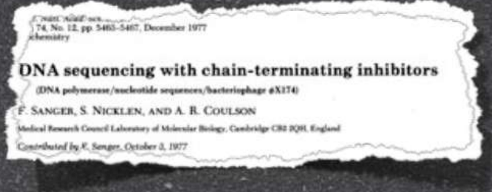


Enzymatic DNA sequencing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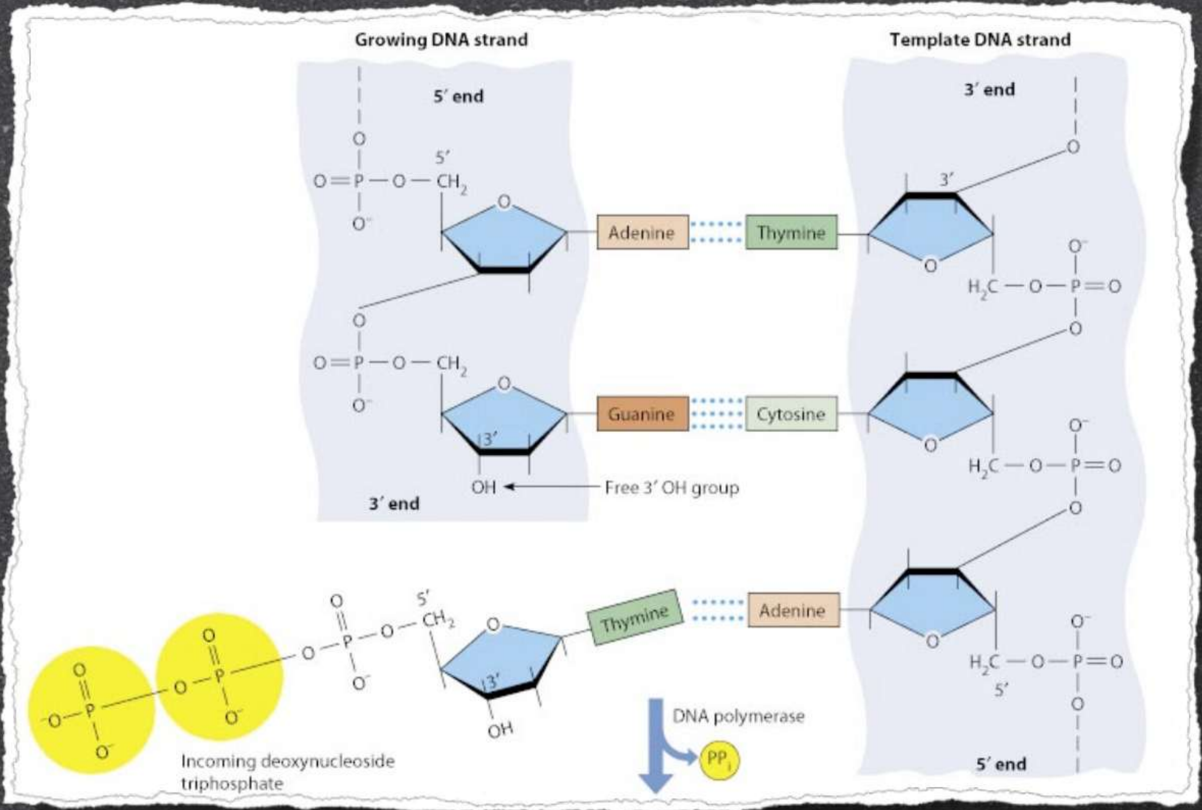

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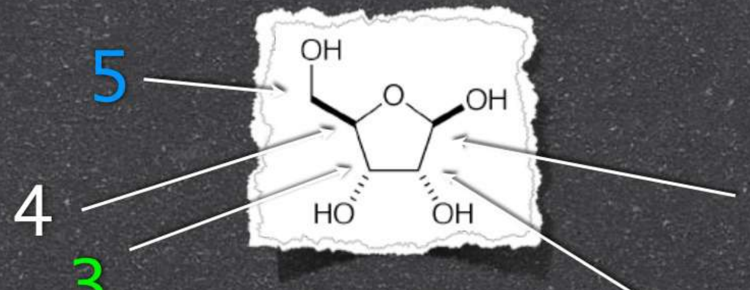
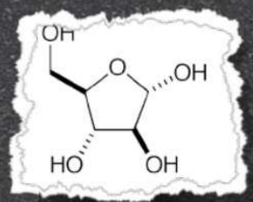
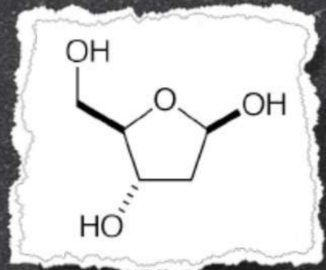
Terminators



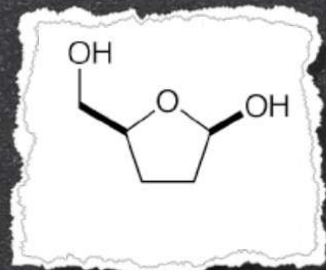
ribose (RNA)



arabinose
(apples)



deoxyribose (DNA)



dideoxyribose

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Enzymatic DNA sequencing

Sanger Recipe

Klenow, Boehringer, Mannheim) (0.2 units). Each mix contained 1.5 × H buffer, 1 μCi of [α-^{32}P]dATP (specific activity approximately 100 mCi/μmol) and the following other triphosphates

phosphates.

ddT: 0.1 mM dGTP, 0.1 mM dCTP, 0.005 mM dTTP, 0.5 mM ddTTP

ddA: 0.1 mM dGTP, 0.1 mM dCTP, 0.1 mM dTTP, 0.5 mM ddATP

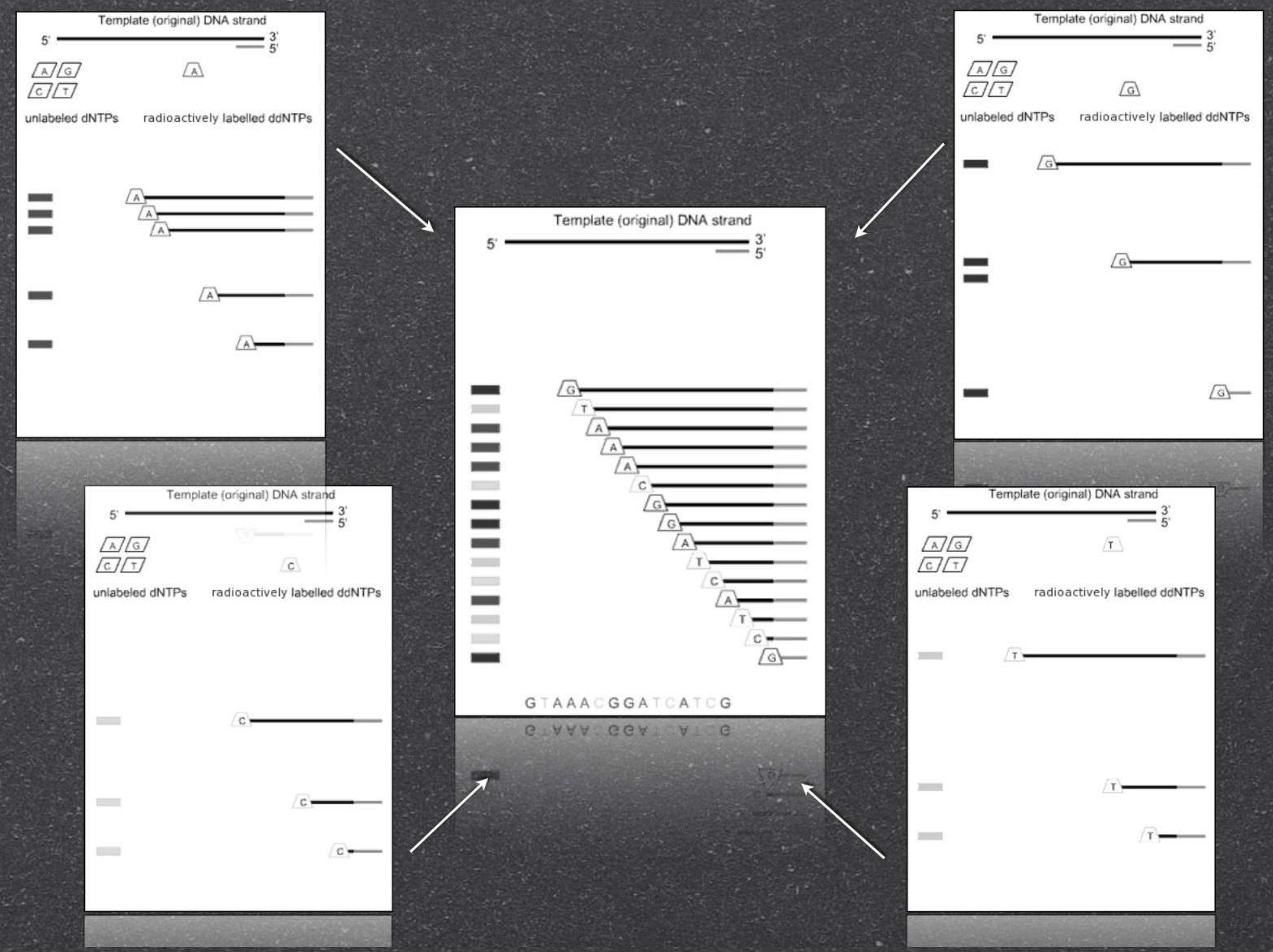
ddG: 0.1 mM dCTP, 0.1 mM dTTP, 0.005 mM dGTP, 0.5 mM ddGTP

ddC: 0.1 mM dGTP, 0.1 mM dTTP, 0.005 mM dCTP, approximately 0.25 mM ddCTP

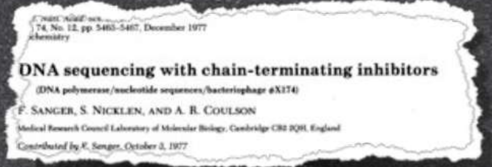
(The concentration of the ddCTP was uncertain because there was insufficient yield to determine it, but the required dilution of the solution was determined experimentally.)

araC: 0.1 mM dGTP, 0.1 mM dTTP, 0.005 mM dCTP, 12.5 mM araCTP

Incubation was at room temperature for 15 min. Then 1 μl



Template (original) DNA strand
5'
3'
5'
A G C T
unlabeled dNTPs
radioactively labelled ddNTPs
A
G
C
T
G T A A A C G G A T C A T C G



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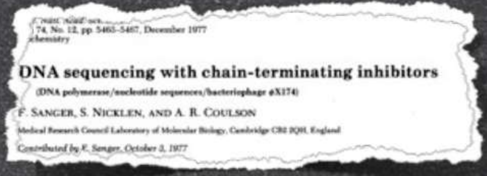

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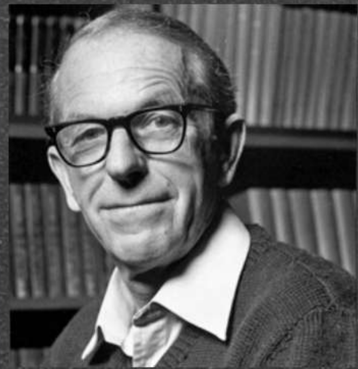

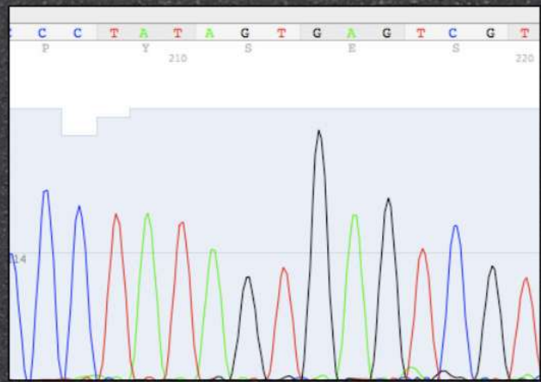


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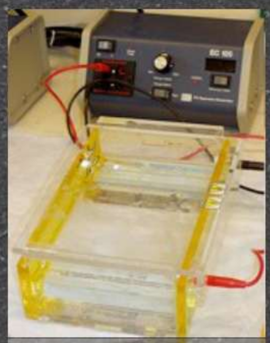
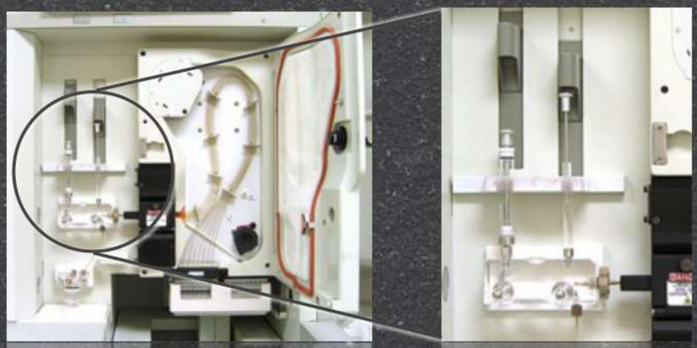
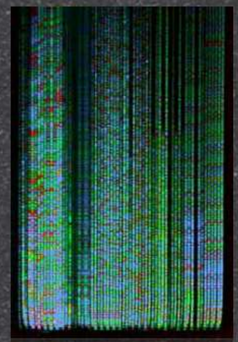
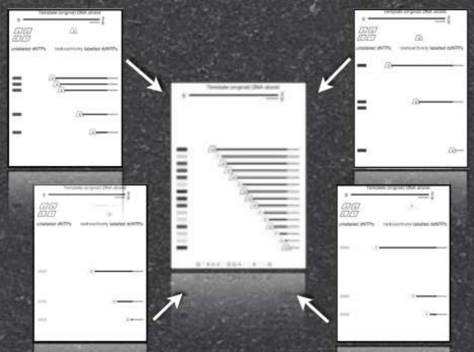
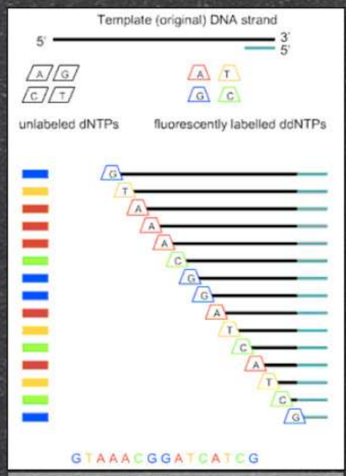






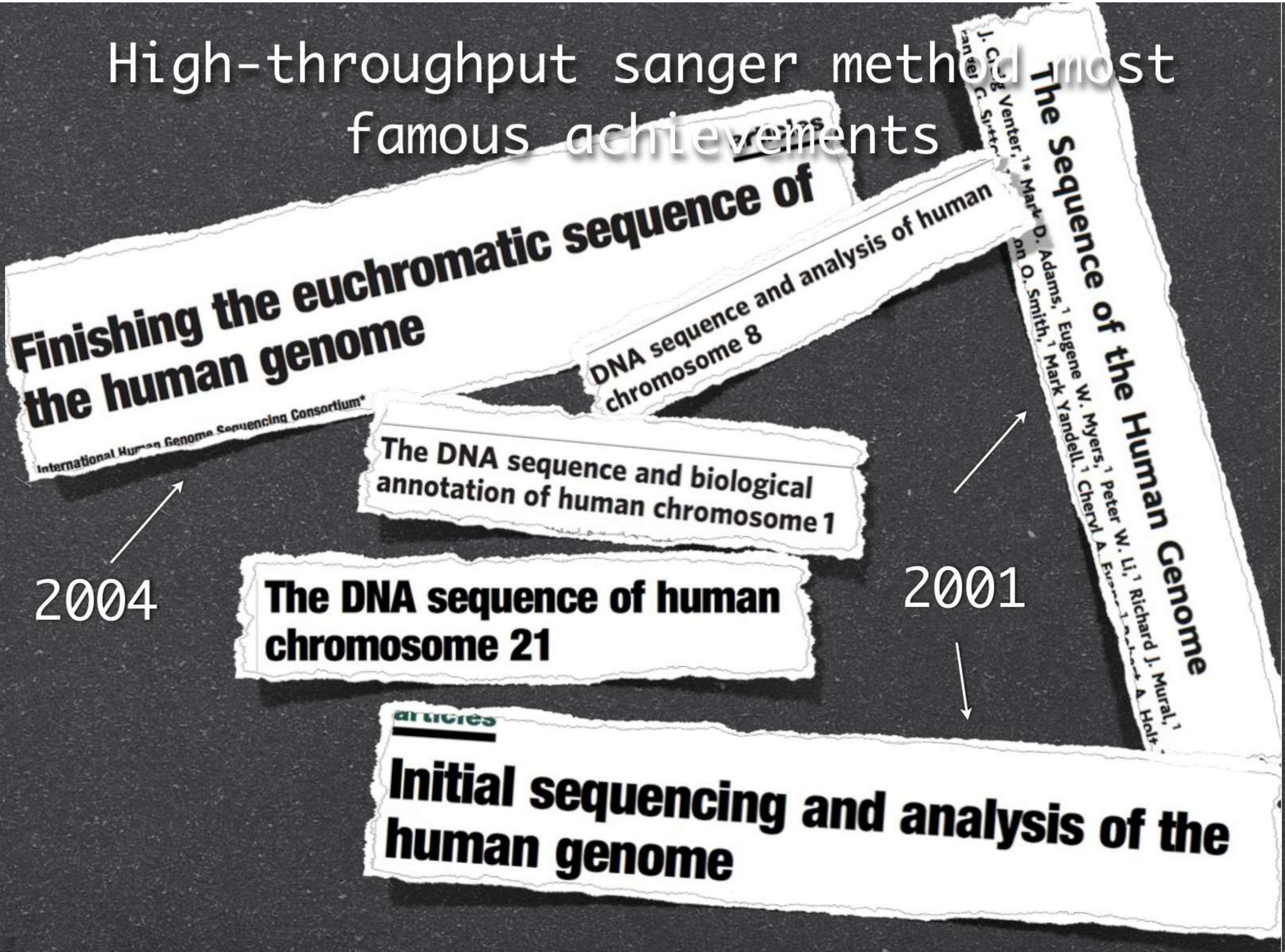




DNA sequencing

Maxam/ Gilbert	Sanger (1977)	Sanger (today)
		Reliable
	Reliable, Easier than chemical sequencing	Provides the longest single reads among all sequencing methods (900-1100 bp)
Does not require any sequence knowledge	Requires partial information about the DNA sequence	Easy-to-handle, one-tube fluorescent labelling; automatable
Radioactive labelling	Radioactive labelling	Requires partial information about the DNA sequence
Resolution limited by polyacrylamide gel	Resolution limited by polyacrylamide gel	Very expensive, not well suited for high-throughput sequencing
Time-consuming	Time-consuming	
Error-prone	Error-prone	

High-throughput sanger method most famous achievements

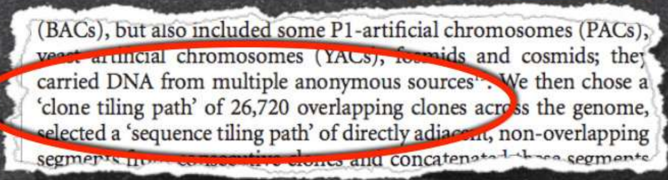


Genome library construction

genome is fragmented in pieces of desired sizes, then cloned into suitable vectors

suitable vectors are chosen depending on the characteristics of the sequences being cloned and sequenced

PACs BACs plasmids fosmids YACs cosmids


(BACs), but also included some P1-artificial chromosomes (PACs), yeast artificial chromosomes (YACs), fosmids and cosmids; they carried DNA from multiple anonymous sources. We then chose a 'clone tiling path' of 26,720 overlapping clones across the genome, selected a 'sequence tiling path' of directly adjacent, non-overlapping segments from consecutive clones and concatenated these segments

each clone is fragmented, then individual fragments are cloned, then sequenced to cope with Sanger sequencing length limits

Shotgun sequencing of each clone

Sequencing whole genomes workflow

Whole genome draft

Sequence assembly

Contig reconstruction

Annotation!

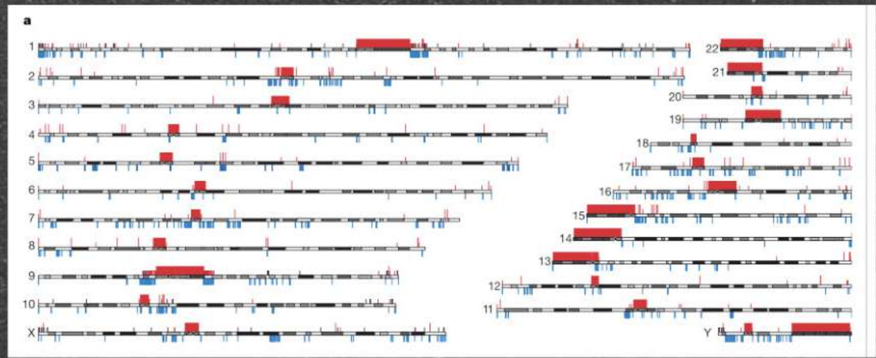
Complexity of such a project

Table 1 **Bases sequenced in Build 35**

Centre	Finished sequence totals (kb)	Finished sequence in Build 35 (kb)
SC	919,388	849,650
WUGSC	645,062	583,032
WIBR	562,096	373,760
JGI/SHGC	485,085	313,988
BCM	320,735	280,963
RIKEN	155,769	112,047
UWGC	145,745	105,573
GS	99,970	78,467
GTC	45,710	32,972
UWMSC	39,227	28,367
Keio	44,905	20,780
IMB	73,677	20,053
Beijing	38,079	17,114
MPIMG	9,838	5,673
GBF	8,325	5,547
UOKNOR	18,657	5,311
TIGR	10,390	3,054
CGM	2,768	1,766
SDSTDC	7,792	1,403
UTSW	8,555	196
Other	30,745	11,621
Total	3,672,516	2,851,336

A number of laboratories has been involved worldwide for 14 years

After 14 years, heterocromatic regions are still refractory to high-throughput-sequencing



Completing the euchromatic sequence is an important goal, but is clearly now a research effort rather than a high-throughput project. Sequencing the human heterochromatin poses an even greater challenge. The current human sequence penetrates only the periphery of the heterochromatin—for example, the pericentric regions on a few chromosome arms[39,40]. This progress has required concerted efforts with specialized mapping techniques and painstaking assembly. The fundamental issue ... current shotgun

Difficult regions must be pproached separatedly

of extensive highly repetitive sequences or near-exact segmental duplication. Telomeric regions of chromosomes were obtained by using a specialized cloning system ('half-YACs vectors') in which successful propagation required that the human insert DNA provided telomeric function. By far, the most difficult regions of the genome were those

5. Roughly 2.19 Mb of sequence on chromosome Y was derived directly from the equivalent pseudoautosomal region on chromosome X.

...ic/centromeric sequences, for which no clone was available (see text).

Next generation sequencing

Roche 454 SEQUENCING

pyrosequencing

Solexa illumina

reversible-terminators sequencing

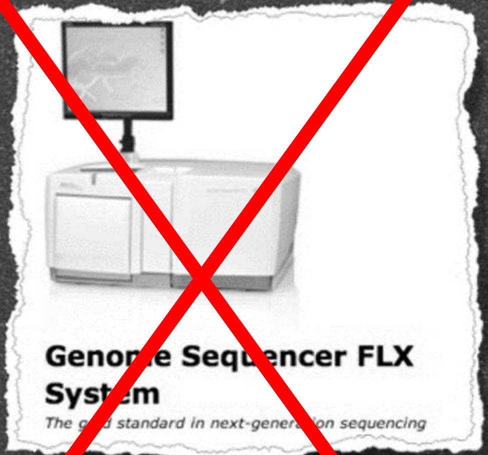
AB applied biosystems SOLiD™

sequencing-by-ligation

back in 2007

Next generation sequencing

Roche 454 SEQUENCING



pyrosequencing

Solexa illumina

reversible-terminators sequencing

2026

applied biosystems SOLiD™

sequencing-by-ligation

Benchtop Sequencers

	MiSeq i100	NextSeq 550	NextSeq 1000/2000
Max output per flow cell	30 Gb	120 Gb	540 Gb
Runtime	4-24 h	11-29 h	8-44 h
Max reads / run *(single)*	100M	400M	1.8G
Max read length	2 × 500 bp	2 × 150 bp	2 × 300 bp

https://www.illumina.com/systems/sequencing-platforms.html

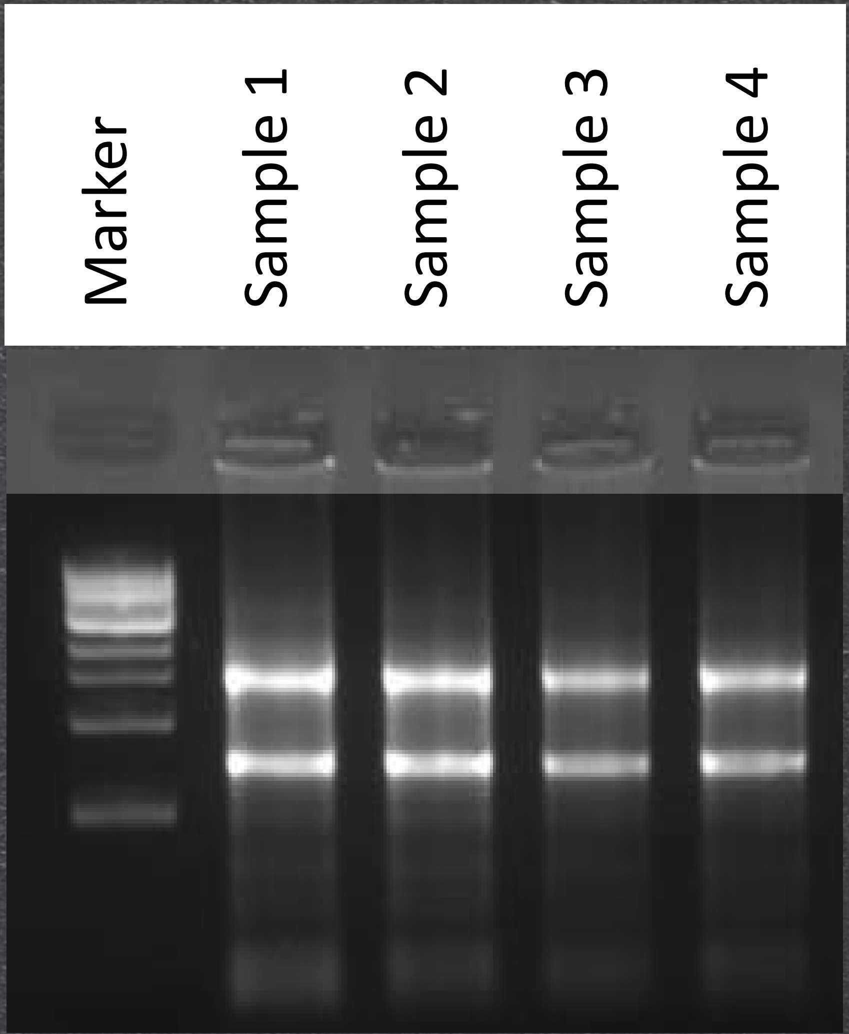
Production-scale sequencers

	NextSeq 1000/2000	NovaSeq 6000	NovaSeq X
			
Max output per flow cell	540 Gb	3 Tb	10.5 Tb
Runtime	8-44 h	13-44 h	17-48 h
Max reads / run *(single)*	1.8G	10/20 B	35/70 B
Max read length	2 × 300 bp	2 × 250 bp	2 × 300 bp

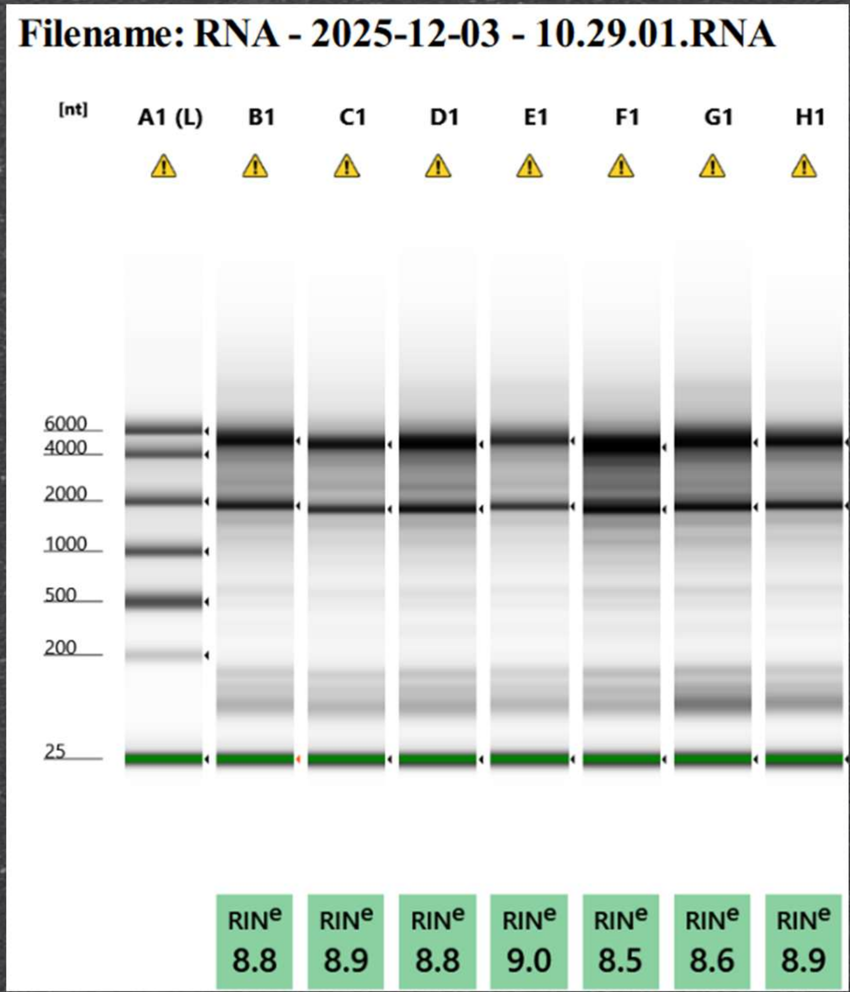
https://www.illumina.com/systems/sequencing-platforms.html

Example workflow

RNA quality



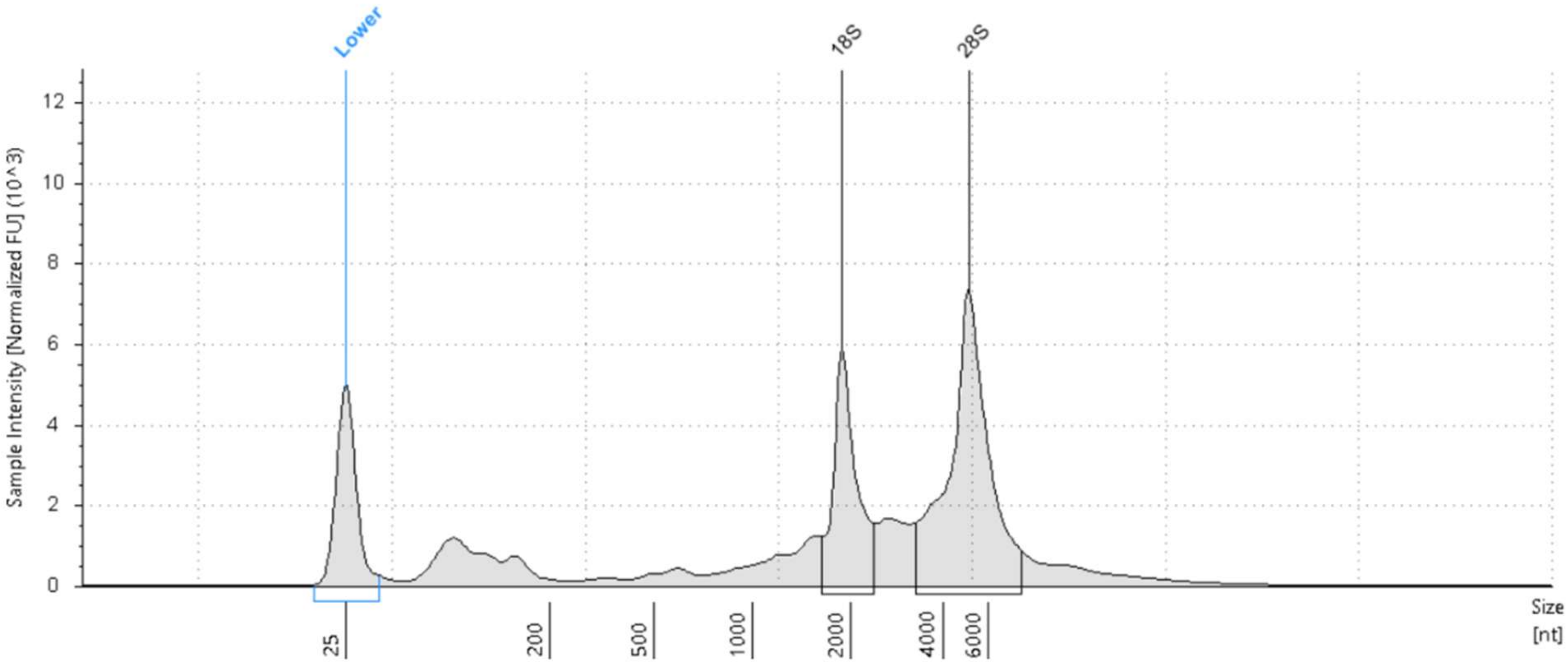
1.7% agarose gel



Agilent TapeStation

RNA quality

B1: A9L4



Sample Table

Well	RINe	Conc. [ng/µl]	Sample Description	Alert	Observations
B1	8.8	137	A9L4	⚠	Caution! Expired ScreenTape device

Illumina workflow – library preparation

1 **Purify and Fragment Messenger RNA**
Hands-on: 46 minutes
Total: 1 hour 21 minutes
Reagents: BBB, BWB, ELB, EPH3, RPBX

2 **Synthesize First Strand cDNA**
Hands-on: 5 minutes
Total: 50 minutes
Reagents: FSA, RVT

3 **Synthesize Second Strand cDNA**
Hands-on: 35 minutes
Total: 1 hour 40 minutes
Reagents: 80% EtOH, AMPure XP, RSB, SMM

4 **Adenylate 3´ Ends**
Hands-on: 5 minutes
Total: 45 minutes
Reagents: ATL4

5 **Ligate Anchors**
Hands-on: 10 minutes
Total: 25 minutes
Reagents: LIGX, RNA Index Anchors, RSB, STL

6 **Clean Up Fragments**
Hands-on: 30 minutes
Total: 30 minutes
Reagents: 80% EtOH, AMPure XP, RSB

7 **Amplify Library**
Hands-on: 10 minutes
Total: 55 minutes
Reagents: EPM, UDP00XX

8 **Clean Up Library**
Hands-on: 25 minutes
Total: 25 minutes
Reagents: 80% EtOH, AMPure XP, RSB

Illumina workflow – library preparation

1 **Purify and Fragment Messenger RNA**
Hands-on: 46 minutes
Total: 1 hour 21 minutes
Reagents: BBB, BWB, ELB, EPH3, RPBX

illumina

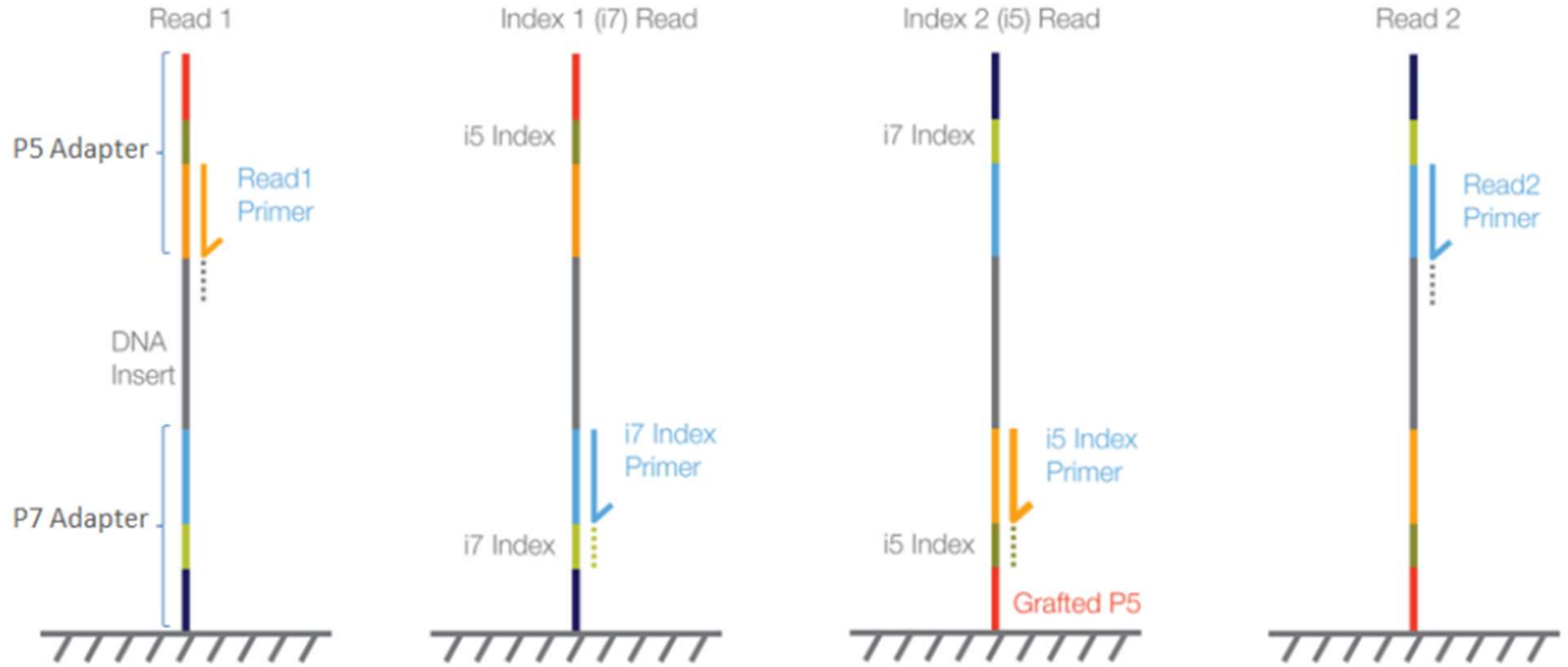
Purify and Fragment mRNA

This step uses oligo(dT) magnetic beads to capture messenger RNAs (mRNAs) with polyA tails. The RNA is then fragmented and primed for complementary DNA (cDNA) synthesis.

Random examers

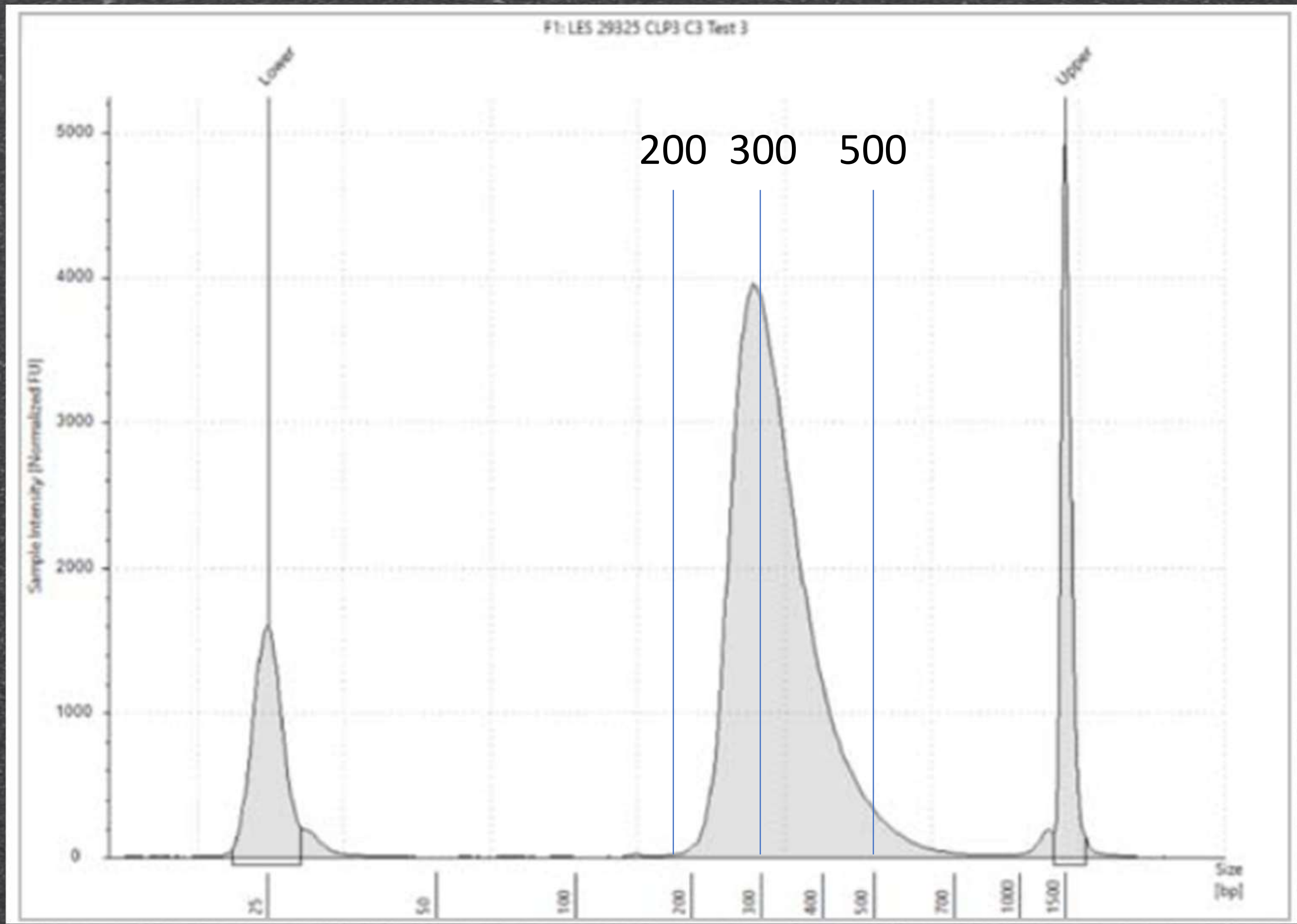
Illumina workflow – library preparation

Figure 2. iSeq 100, MiniSeq, NextSeq 500/550, NextSeq 1000/2000, and NovaSeq 6000 (with v1.5 reagents)



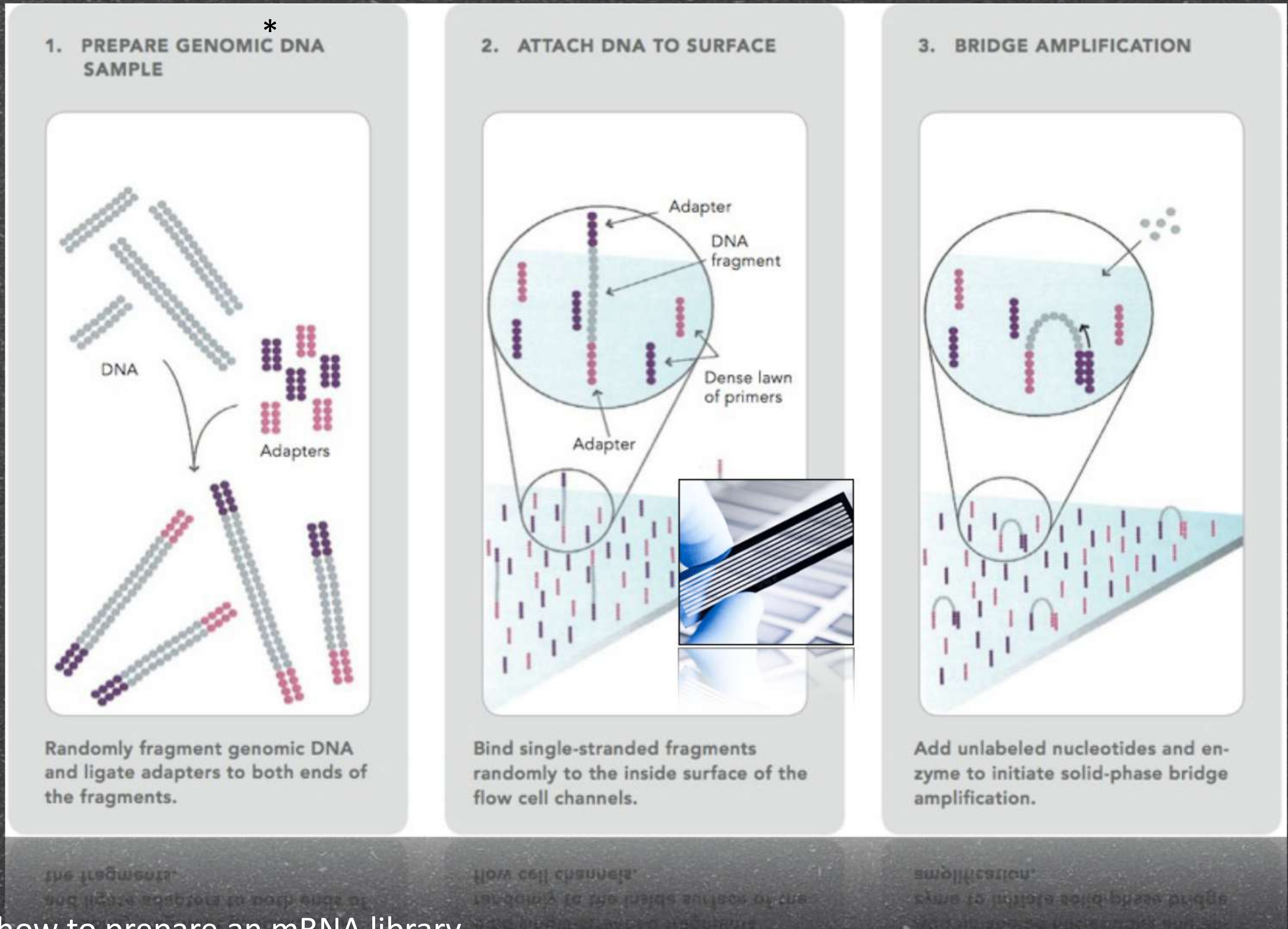
https://knowledge.illumina.com/software/general/software-general-reference_material-list/000002905

Illumina workflow – library preparation



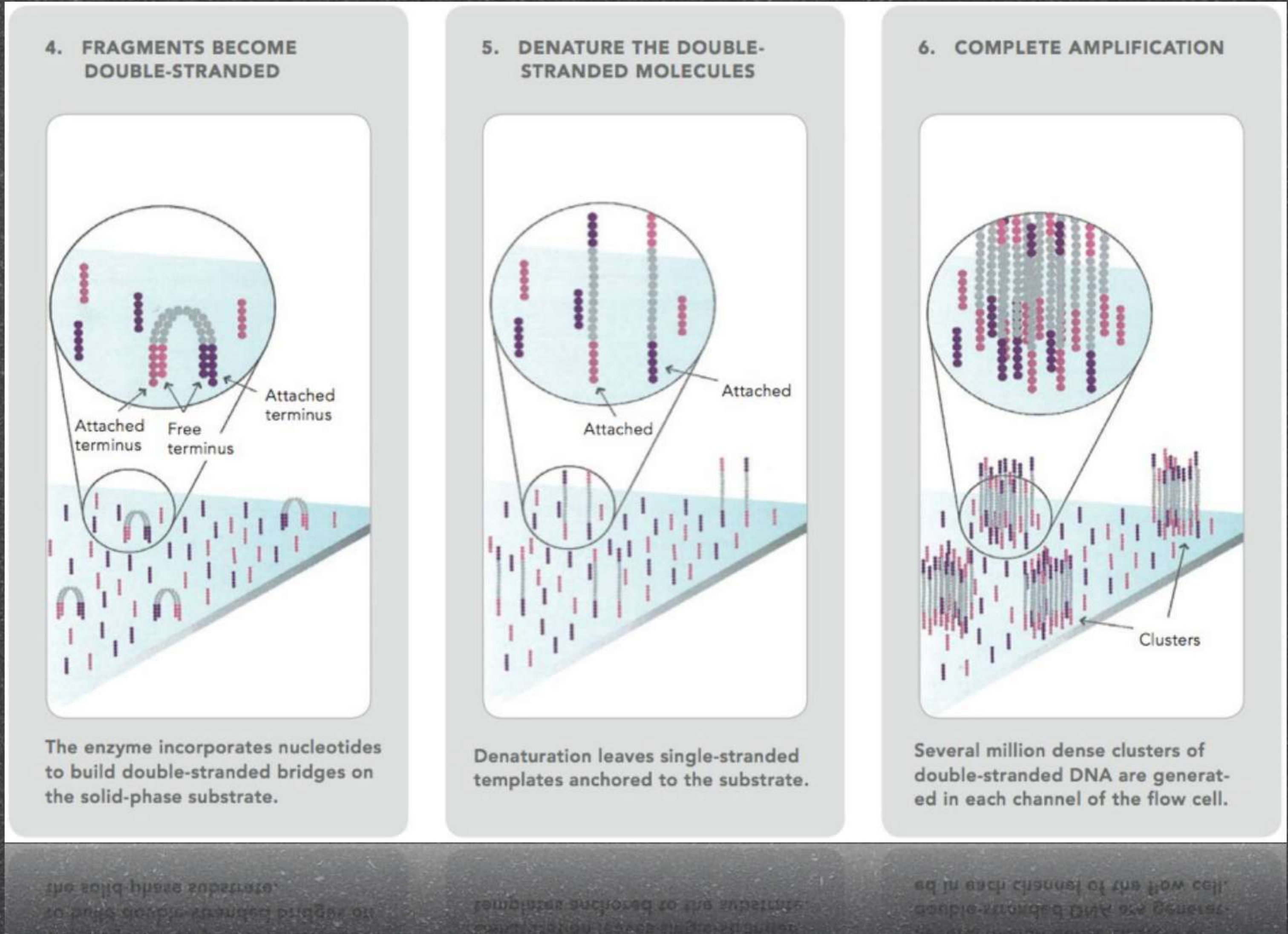
The expected insert size depends on the read length

Illumina workflow

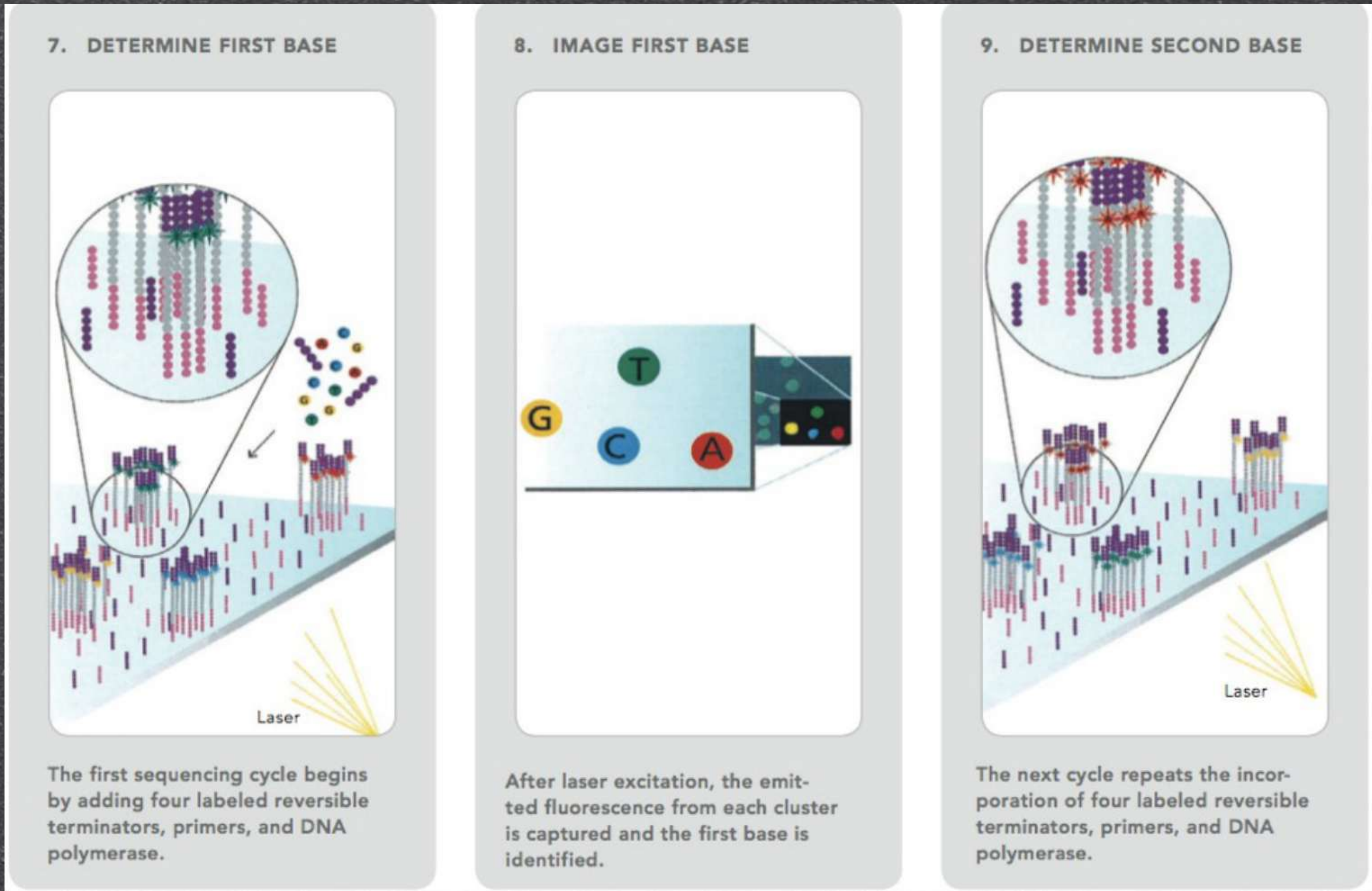


* we already saw how to prepare an mRNA library

Illumina workflow



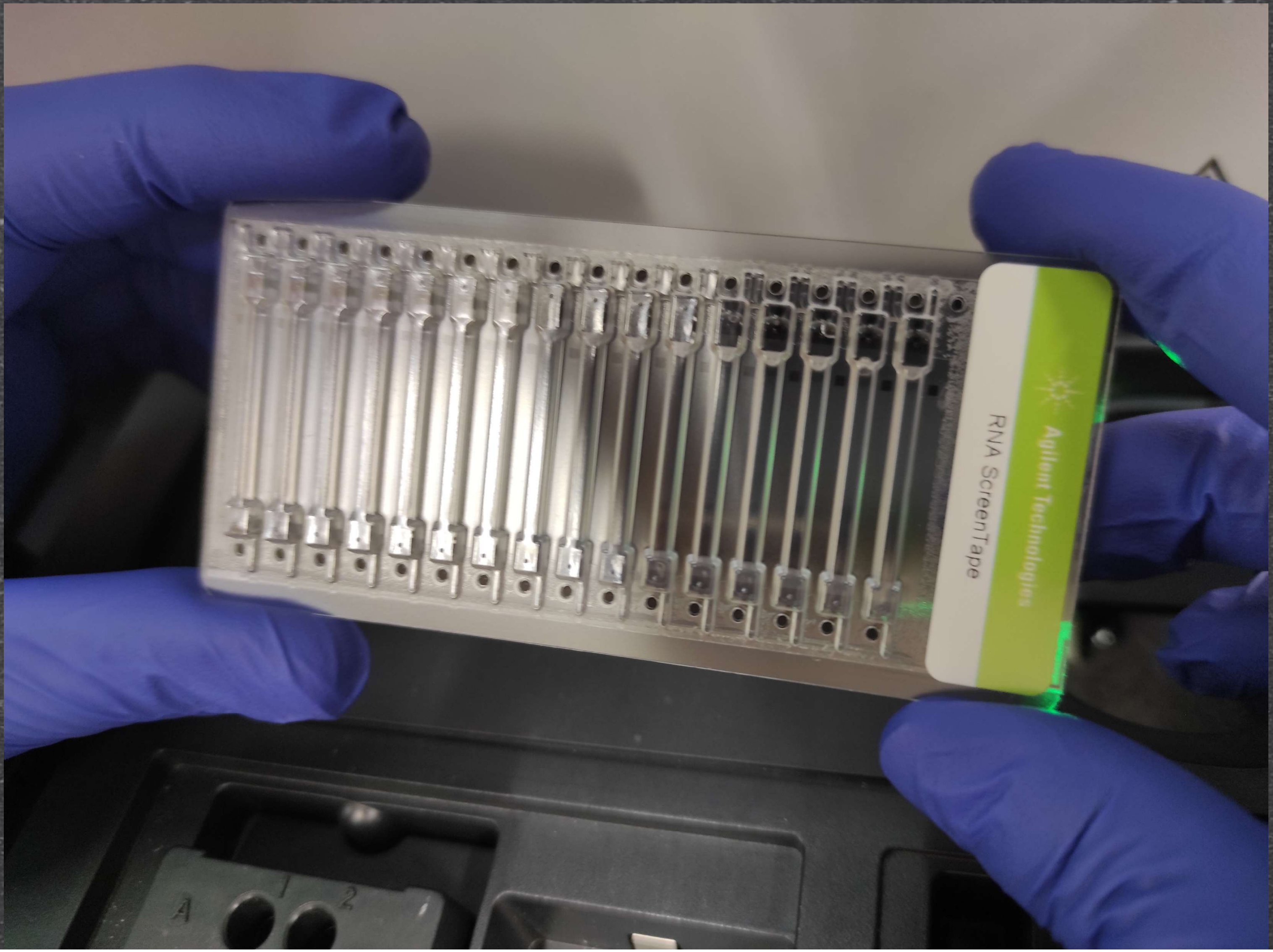
Illumina workflow



Tape station

Tape station

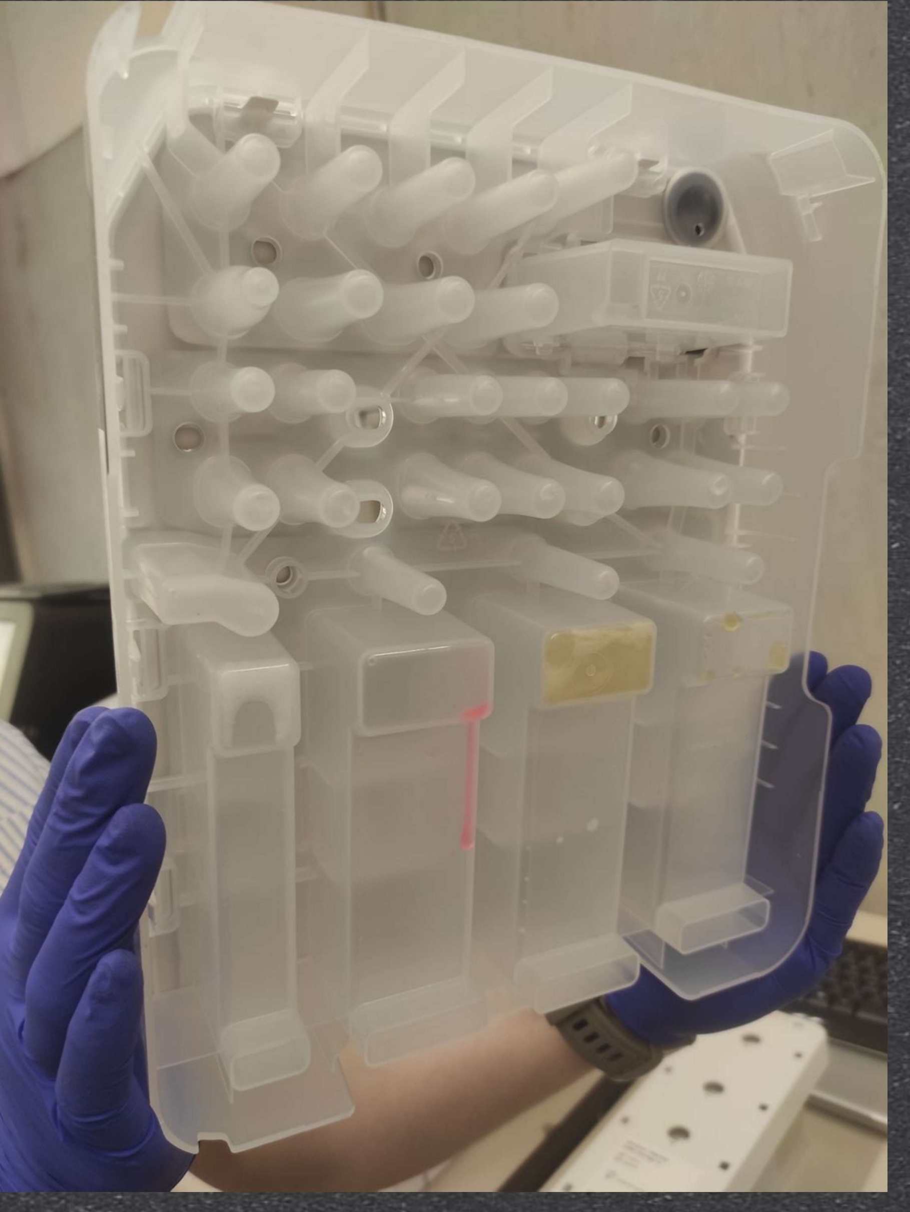
Tape station: screentape



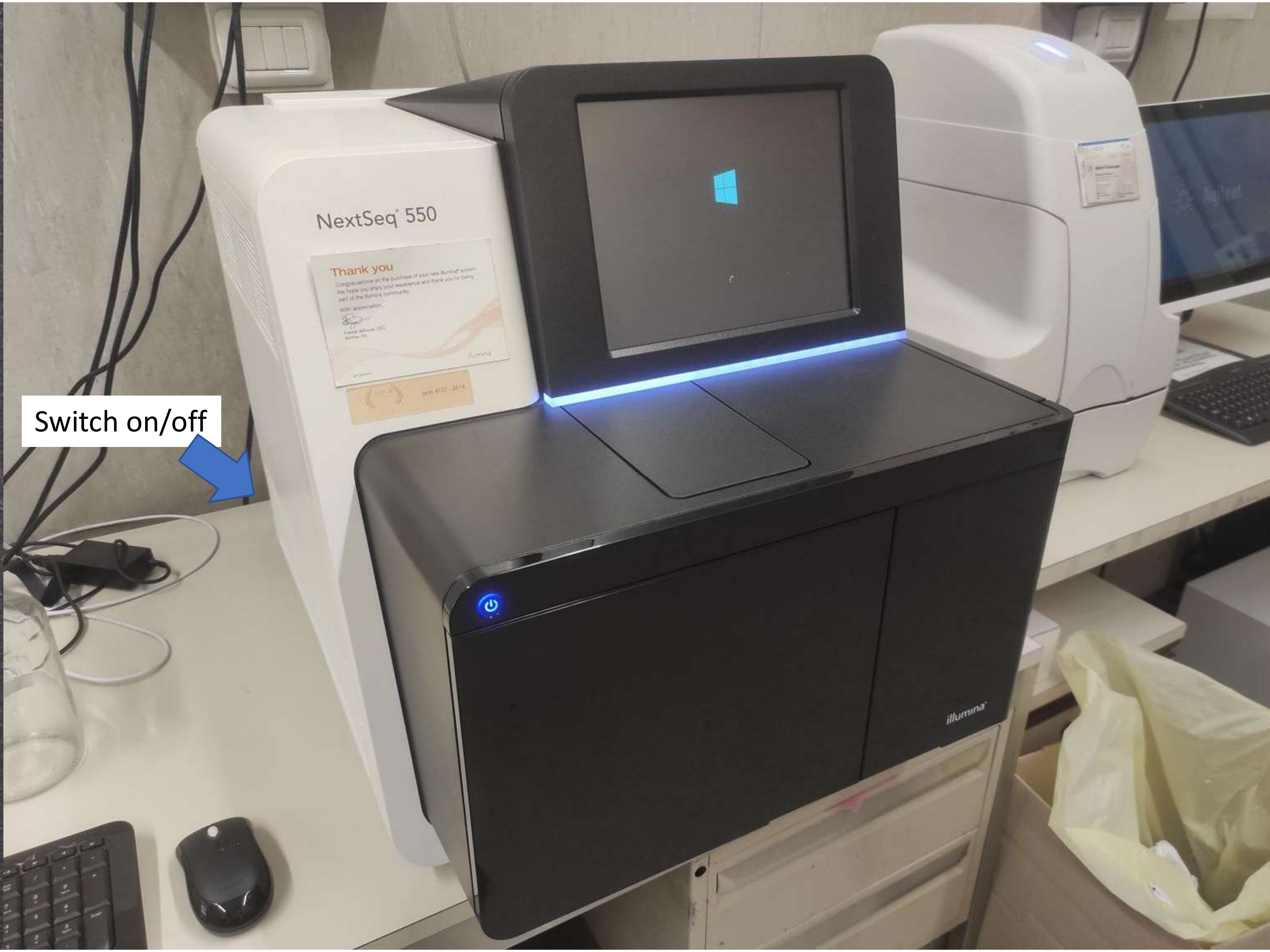
Tape station: screentape

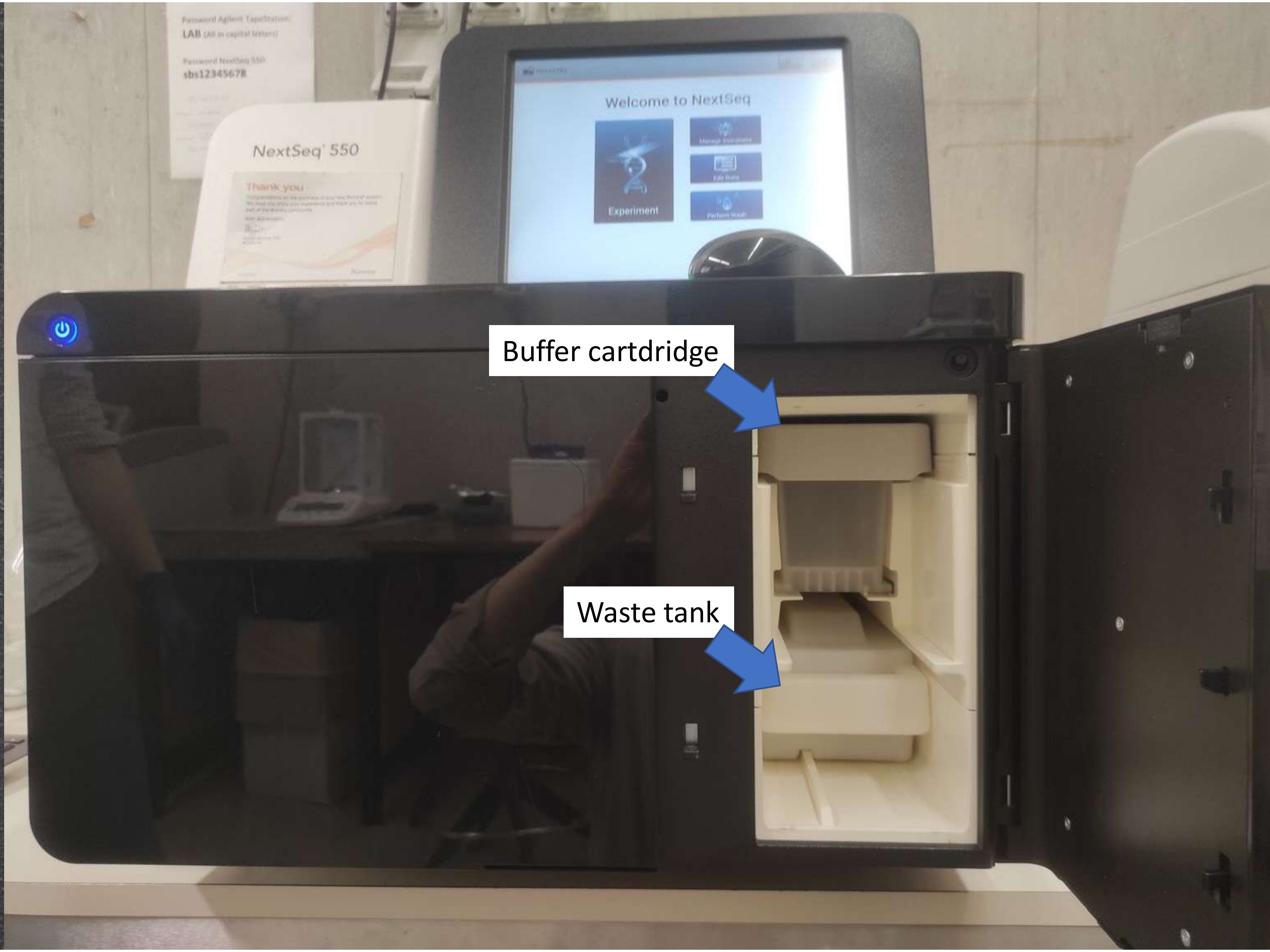
NextSeq 550: Flow cell

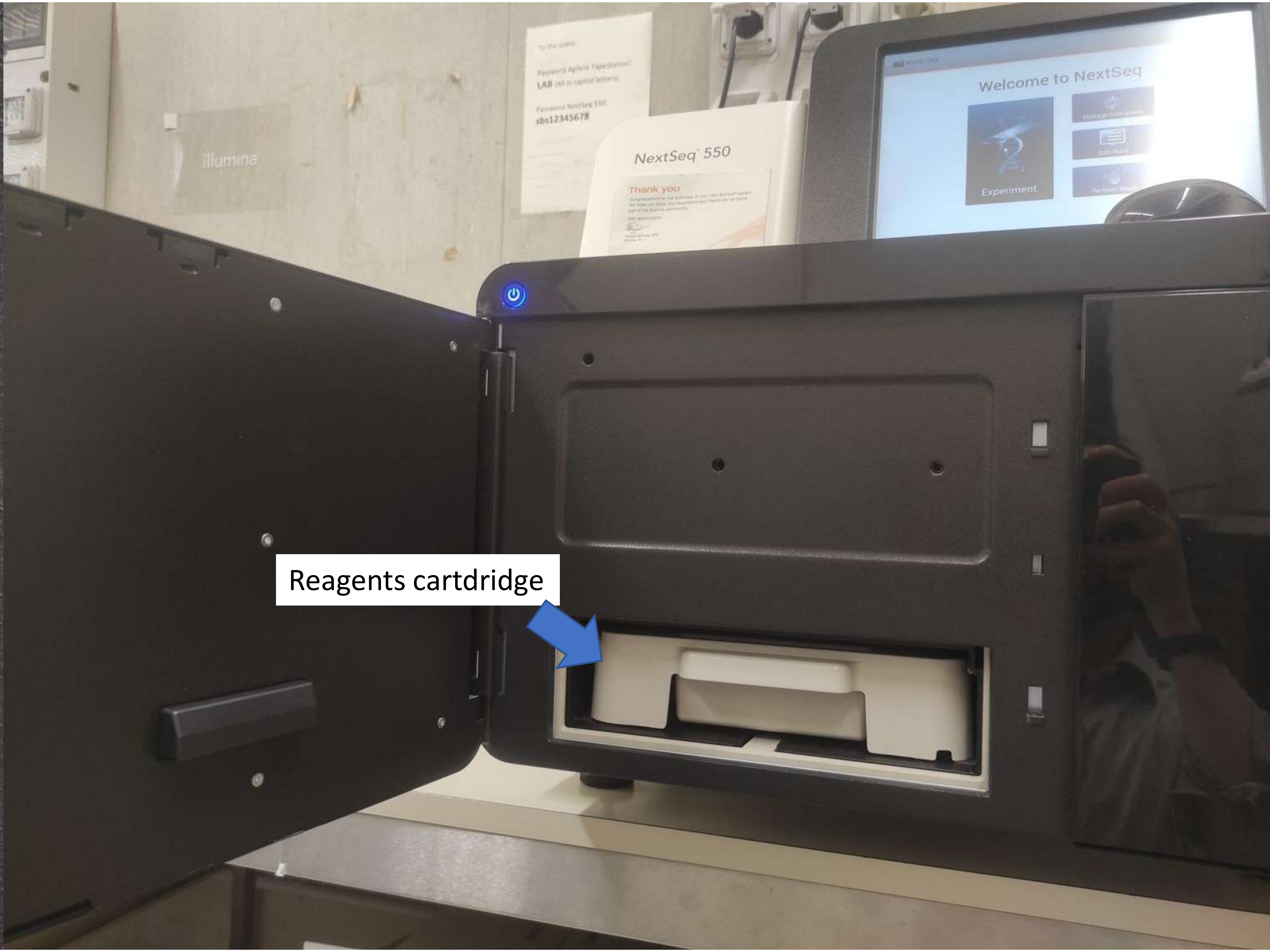
Insert
Load Library Here
Detach Position 6 Here
NextSeq™ 500/550
High Output Reagent Cartridge v2
HIGH
version v2
cycles 150
REF 15057931
LOT 20937590
STORE IN THE DARK
illumina



NextSeq 550: Buffer cartridge


Switch on/off
NextSeq 550
Thank you
illumina


Buffer cartdridge
Waste tank
NextSeq 550
Welcome to NextSeq
Experiment
Thank you
Password Agilent TapeStation:
LAB (All in capital letters)
Password NextSeq 550:
sbs12345678


Reagents cartdridge
Welcome to NextSeq
Experiment
NextSeq 550
Thank you
illumina


NextSeq 550
Thank you
NB552283
Welcome to NextSeq
Experiment
Manage Instrument
Edit Runs
Perform Wash
Flow cell compartment

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